

$$\begin{aligned} \cdot \operatorname{ch}^2(x) - \operatorname{sh}^2(x) &= 1 \\ \cdot \operatorname{ch}(x+y) &= \operatorname{ch}(x)\operatorname{ch}(y) + \operatorname{sh}(x)\operatorname{sh}(y) \\ \cdot \operatorname{ch}(x-y) &= \operatorname{ch}(x)\operatorname{ch}(y) - \operatorname{sh}(x)\operatorname{sh}(y) \\ \cdot \operatorname{sh}(2x) &= 2\operatorname{sh}(x)\operatorname{ch}(x) \\ &= 2\operatorname{ch}^2(x) - 1 \\ &= 1 - 2\operatorname{sh}^2(x) \end{aligned}$$

$$\begin{aligned} \cdot \operatorname{sh}(x+y) &= \operatorname{sh}(x)\operatorname{ch}(y) + \operatorname{ch}(x)\operatorname{sh}(y) \\ \cdot \operatorname{sh}(x-y) &= \operatorname{sh}(x)\operatorname{ch}(y) - \operatorname{ch}(x)\operatorname{sh}(y) \end{aligned}$$

$$\operatorname{sh}(2x) = 2\operatorname{sh}(x)\operatorname{ch}(x)$$

$$\operatorname{ch}(x+y) = \frac{1 + \operatorname{ch}(x)\operatorname{ch}(y)}{\operatorname{ch}(x) + \operatorname{ch}(y)}$$

$$\operatorname{ch}(x-y) = \frac{1 - \operatorname{ch}(x)\operatorname{ch}(y)}{\operatorname{ch}(x) - \operatorname{ch}(y)}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \operatorname{Argsh}(x) &= \ln(x + \sqrt{x^2 + 1}) \\ \lim_{x \rightarrow -\infty} \operatorname{Argsh}(x) &= -1 \quad \operatorname{Argsh}(\operatorname{sh}(x)) = x \quad \forall x \in \mathbb{R} \end{aligned}$$

$$x \rightarrow \operatorname{Argsh}(x) = \ln(x + \sqrt{x^2 + 1})$$

$$D_f = \mathbb{R} / \text{com et deriv dans } \mathbb{R}$$

$$\begin{aligned} \operatorname{ch}(\operatorname{Argsh}(y)) &= y \quad \forall y \in [1, +\infty[\\ \operatorname{Argsh}(\operatorname{ch}(x)) &= x \quad \forall x \in [0, +\infty[\end{aligned}$$

$$x \rightarrow \operatorname{Argsh}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

$$D_f \rightarrow]-1, 1[\text{ contin et deriv dans }]-1, 1[$$

$$\begin{aligned} \lim_{x \rightarrow 1} \operatorname{Argsh}(x) &= +\infty \\ \lim_{x \rightarrow -1} \operatorname{Argsh}(x) &= -\infty \\ \operatorname{Argsh}'(x) &= \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

Primitives

$$\begin{aligned} u' e^u &\rightarrow e^u + C \\ u' \sin u &\rightarrow -\cos u + C \\ u' \cos u &\rightarrow \sin u + C \\ u' \ln u &\rightarrow \frac{1}{u} + C \\ u' \frac{1}{u} &\rightarrow \ln|u| + C \\ u' \operatorname{sh} u &\rightarrow \operatorname{ch}(u) + C \\ u' \operatorname{ch} u &\rightarrow \operatorname{sh}(u) + C \end{aligned}$$

Intégrale

$$\begin{aligned} \int_a^b f(x) dx &= F(b) - F(a) \\ \text{Si } u(x) &= f(x) \text{ dx} = [u]_a^b - \int_a^b u'(x) \operatorname{ch}(x) dx \\ \text{Si } u(x) &= f(x) \text{ dx} = [u]_a^b - \int_a^b u'(x) \operatorname{sh}(x) dx \end{aligned}$$

$$\operatorname{Argsh}'(x) = \frac{1}{\sqrt{x^2 + 1}}$$

Developpement de Taylor

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + o(x^n) \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2}) \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^m \frac{x^{2m}}{(2m)!} + o(x^{2m+1}) \\ \operatorname{sh}(x) &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2}) \\ \operatorname{ch}(x) &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2m}}{(2m)!} + o(x^{2m+1}) \\ \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{m-1} \frac{x^m}{m} + o(x^m) \end{aligned}$$

Si $u_n \geq 0$ et $\sum u_n$ sont de même nature
 $\sum u_n$ (Riemann et Riemann) développement de Taylor